Problem Set 1

1.) The mean of a function over a distribution function is defined as:

$$\langle G \rangle = \frac{\int G(\vec{v}) f(\vec{v})\, d^3v}{\int f(\vec{v})\, d^3v} \qquad \underset{\text{maxwellian}}{f(\vec{v})} = n\left(\frac{m}{2\pi T}\right)^{3/2} \exp\left(-\frac{mv^2}{2T}\right)$$

Note that the maxwellian distribution is isotropic.

a.) Prove that f is correctly normalized:

$$\int f\, d^3v = n$$

Now, $f(\vec{v}) = n\left(\frac{m}{2\pi T}\right)^{3/2} \exp\left(-\frac{mv^2}{2T}\right)$

$$\int f\, d^3v = \int n\left(\frac{m}{2\pi T}\right)^{3/2} \exp\left(-\frac{mv^2}{2T}\right) d^3v = \left(\frac{2\pi T}{m}\right)^{3/2}\left(\frac{m}{2\pi T}\right)^{3/2} n = \underline{\underline{n}} \checkmark$$

↳ Gaussian integral

b.) Find $\langle v_x \rangle$

$$\text{Now } \langle v_x \rangle = \frac{\int v_x f(\vec{v})\, d^3v}{\int f(\vec{v})\, d^3v} = \int\left(\frac{m}{2\pi T}\right)^{1/2} v_x \exp\left(-\frac{mv_x^2}{2T}\right) \int\left(\frac{m}{2\pi T}\right) \exp\left(-\frac{m(v_y^2+v_z^2)}{2T}\right) dv_x\, dv_z\, dv_y$$

$$= \left(\frac{m}{2\pi T}\right)^{1/2} \int v_x \exp\left(-\frac{mv_x^2}{2T}\right) dv_x$$

$$= \frac{1}{2}\left(\frac{m}{2\pi T}\right)^{1/2} \int_{\infty}^{\infty} \exp\left(-\frac{mt}{2T}\right) dt = \underline{\underline{0}} \checkmark$$

c.) $\langle v^2 \rangle$ and therefore $\langle \frac{1}{2}mv^2 \rangle$

$$\text{Now, } \langle v^2 \rangle = \frac{\int v^2 f(\vec{v})\, d^3v}{\int f(\vec{v})\, d^3v} = \int v^2 \left(\frac{m}{2\pi T}\right)^{3/2} \exp\left(-\frac{mv^2}{2T}\right) d^3v$$

$$\text{Now, } \left(\frac{m}{2\pi T}\right)^{1/2} \int v_x^2 \exp\left(-\frac{mv_x^2}{2T}\right) dv_x = \left(\frac{m}{2\pi T}\right)^{1/2}\left[v_x^2 \int \exp\left(-\frac{mv_x^2}{2T}\right) dv_x - \int dv_x\, v_x \cdot \int \exp\left(-\frac{mv_x^2}{2T}\right) dv_x\right]$$

$$= \left(\frac{m}{2\pi T}\right)^{1/2}\left(\frac{2\pi T^3}{m^3}\right)^{1/2} = \frac{T}{m}$$

The triple integral then gives us: $\langle v^2 \rangle = \frac{3T}{m}$ $\quad \left\langle \frac{1}{2}mv^2 \right\rangle = \underline{\underline{\frac{3}{2}T}} \checkmark$

d.) Find $\langle |v| \rangle$

$$\text{Now, } \langle |\vec{x}| \rangle = \langle r \rangle = 4\pi \frac{\int r^3 f(\vec{r})\, dr}{\int f(\vec{x})\, d^3x}$$

Now, $f(r) = n\left(\frac{m}{2\pi T}\right)^{3/2} \exp\left(-\frac{mr^2}{2T}\right)$

$$\therefore \langle r \rangle = 4\pi \left(\frac{m}{2\pi T}\right)^{3/2} \int_0^\infty r^3 \exp\left(-\frac{mr^2}{2T}\right) dr = 4\pi \left(\frac{m}{2\pi T}\right)^{3/2} \left(\frac{2T^2}{m^2}\right) = \left(\frac{8T}{m\pi}\right)^{1/2} \checkmark$$

2.) Show that for any isotropic distribution function one can integrate over particles having positive velocity component along one axis to prove that the flux density with which particles impinge upon a surface from one side only is $\frac{1}{4} n \langle |v| \rangle$.

Consider particles incident like this:

Here the velocity is only in one (true) dir^n so to calculate the flux, there is no need to account for the negative velocities since at this boundary by definition there are none.

$\therefore$ Flux density = Number density x No. of particles with velocity in that dir^n

$$= n \times \int_0^\infty |v| f(|v|) d|v| = n \times \left(\frac{m}{2\pi T}\right)^{1/2} \times \left(\frac{T}{m}\right) = n \times \left(\frac{T}{2\pi m}\right) = \frac{n}{4}|v| \checkmark$$

3.) Suppose the degree of ionization of a gas discharge is governed by the Saha equation:

$$\frac{n_e n_i}{n_0} = 2\frac{(2\pi m_e T)^{3/2}}{h^3} \exp\left(-\frac{\chi_i}{T}\right) \quad \chi_i \to 13.6\,eV$$

Calculate the temperature approximately at which the gas is 50% ionized. The total pressure of the gas = 1atm. What then is the ratio of the electron density to the density of a room temperature gas at the same pressure.

$n_i = n_e$; $n_i = \frac{n_0}{2}$

$$\therefore \text{we get: } \frac{(n_0/2)^2}{n_0} = 2\frac{(2\pi m_e T)^{3/2}}{h^3} \exp\left(-\frac{\chi_i}{T}\right) \quad \text{where } n_0 = \frac{RT}{P_{1atm}}$$

We can use the above to calculate T.

At room temp. $n_i \neq n_e$, $n_i = \frac{n_0}{2}$, $n_0 = \frac{RT}{P_{1atm}}$

$$\therefore \frac{n_e}{n_0} = \frac{4(2\pi m_e T)^{3/2}}{h^3} \exp\left(-\frac{\chi_i}{T}\right) \times \frac{P}{RT} \quad \to \text{Proceed from here :)} \checkmark$$